Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

***Benchmark:** CRISISLTLSUM* | ***Context:** Mexico City and Central America Bilingual Crisis Journalism*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All input content was collected from seven US states and two Australian states during 2020–2021 [Q123], using English-language keyword lists curated from English news and social media [Q126, Q127], English Twitter NER models [Q35], and OpenStreetMap location augmentation anchored to US/Australian gazetteers [Q30, Q36]. The authors themselves note that replication for other languages or regions would be difficult [Q119]. For the target deployment, no Mexican or Central American content is present; Mexican administrative geography (16 alcaldías, colonias, 32 estados, municipios) and Central American departamentos/municipios are entirely absent. The web research confirms that no Spanish-language crisis tweet corpus anchored to Mexican or Central American events exists [WEB-3, WEB-4, WEB-6]. Mexican Spanish Twitter conventions (diacritic dropping, regional vocabulary like 'temblor'/'sismo'/'edomex', SASMEX hashtags #TenemosSismo, voseo in Central America, code-switching at the border) are entirely unrepresented in the benchmark's content [WEB-7, WEB-8]. The keyword-filtering pipeline is also acknowledged as not comprehensive [Q132], and its effectiveness depends on English-curated lists [Q128].

ID	Evidence
Q30	“The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest. A tweet is considered relevant to our area of interest if 1) the text mentions one of the candidates, 2) the geo-tag matches the area of interest, or 3) the user location matches the area of interest.” (p.3)
Q35	“Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions.” (p.4)
Q36	“Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses.” (p.4)
Q119	“Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies.” (p.10)
Q123	“We collect CrisisLTLSum across seven states in the USA and two states of Australia during 2020 and 2021. The selected states in the USA are New York, Illinois, Massachusetts, California, Texas, and Florida. From Australia, we have selected the states of New South Wales and Victoria.” (p.12)
Q126	“We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past.” (p.13)
Q127	“Please note that these lists are generic for each category and not specific to unique events. For instance, instead of defining keywords specific to an event called "Hurricane Ida", our keywords list includes phrases such as "fallen tree", "building collapse", or "storm".” (p.13)
Q128	“The quality of the keywords list is crucial to the final quality of the generated timelines, and we polish this list multiple times based on small sets of experiments before using them for the final task.” (p.13)
Q132	“This approach will not be comprehensive or exhaustive but rather representative of each crisis domain. Improving this method to be more encompassing is an area for future research.” (p.13)
WEB-3	HumAID is English-only despite including a 2017 Mexico earthquake subset — confirming the gap in Spanish-language Mexican crisis tweet content (https://crisisnlp.qcri.org/humaid_dataset.html)
WEB-4	Curiel et al. (PMC 2019) is the only known Mexican-Spanish crisis NER/geolocation resource and is not a timeline/summarization dataset (https://pmc.ncbi.nlm.nih.gov/articles/PMC6484392/)
WEB-6	https://github.com/wangcongcong123/crisis_nlp_progress/blob/master/datasets/TREC-IS.md
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)

WEB-8	@SASMEX X account provides structured earthquake seed tweets with no analog in benchmark content (https://x.com/sasmex)

Strengths

- The general principle of geographic and temporal focus (specific area × domain × time period) [Q29, Q32] is methodologically transferable, even though the specific filters used are US-anchored.
- The benchmark explicitly acknowledges its non-replicability for other languages/regions [Q119], providing transparency that supports informed adaptation decisions.

ID	Evidence
Q29	"We limit the incoming tweets to specific geographical areas, periods, and domains of interest." (p.3)
Q32	"The combinations of (area a, domain d, time t) are manually selected so that the events of type d at location a are more frequent during time period t." (p.3)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — target inputs require Mexican Spanish dialectal knowledge, alcaldía/colonia/municipio toponymy, Central American voseo, indigenous-language place names (Nahuatl, Zapotec, Maya, Mam, K'iche'), and SASMEX/CENAPRED hashtag conventions, none of which are present in benchmark content [Q123, WEB-4, WEB-7].

IC-2: No — the benchmark's English-language US/Australian content does not align culturally or linguistically with bilingual Mexican/Central American crisis journalism [Q123, Q126].

IC-3: Yes — input content is filtered through English-language keyword lists [Q126, Q127] and US/Australian location lists [Q30] that would not transfer to Spanish-language Mexican crisis tweets. The OpenStreetMap-based location augmentation [Q36] requires Spanish-language gazetteer adaptation not provided.

IC-4: INSUFFICIENT DOCUMENTATION — no Spanish-speaking annotators or regional reviewers were involved in content selection [Q52, Q61]. Recruitment of Mexican/Central American journalists or linguists would be required to identify culturally sensitive instances.

IC-5: Content validity is severely compromised: the benchmark contains zero Spanish-language content, zero Mexican/Central American geographic anchors, and uses an English-only NER + OSM pipeline that the authors explicitly note is not portable [Q119].

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Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q52	"We, authors of this work, manually selected 1,000 clusters that contain enough tweets describing how a crisis event evolves, while specifying the "seed tweet" (i.e. the first observed post that describes the ongoing event) of each timeline." (p.5)

Q61	“First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found.” (p.5)
Q119	“Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies.” (p.10)
Q123	“We collect CrisisLTLSum across seven states in the USA and two states of Australia during 2020 and 2021. The selected states in the USA are New York, Illinois, Massachusetts, California, Texas, and Florida. From Australia, we have selected the states of New South Wales and Victoria.” (p.12)
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WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)

Information Gaps

- Frequency of indigenous-language place names in crisis tweets unknown
- Voseo penetration in Central American crisis tweets unquantified

Requires Expert Verification

- Mexican/Central American journalist review of which colonia/municipio/landmark references are most frequent in crisis tweet streams
- Linguist review of code-switching patterns at US-Mexico border

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Remediation

Gap 1: Zero Spanish-language Mexican/Central American content; English NER + English-OSM pipeline is non-portable [Q35, Q36, Q119]

Recommendation: Build or commission a Spanish-language crisis tweet corpus anchored to Mexican (CDMX alcaldías, key estados) and Central American (departamentos/municipios) events. Adapt the Curiel et al. (PMC 2019) Mexican-Spanish NER pipeline [WEB-4] for georesolution, and integrate a Spanish-language gazetteer that includes colonias, alcaldías, and indigenous-language place names.

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Q35	“Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions.” (p.4)
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